Reinforcement Learning on ALE/KungFuMaster

A Comparative Study of DQN and PPO

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Outline

Introduction and Motivation

Project Goal

Compare two foundational families of deep RL algorithms on a challenging Atari benchmark to understand their relative strengths, sample efficiency, and behavior under custom reward shaping.

Why DQN vs PPO?

- **DQN** (Mnih et al., 2015): off-policy, value-based.
- **PPO** (Schulman et al., 2017): on-policy, policy-gradient with clipped surrogate.
- Two dominant deep-RL paradigms.

Why ALE/KungFuMaster-v5?

- Sparse, deceptive rewards (reward-hacking risk).
- Long-horizon credit assignment across hundreds of steps.
- Visual complexity (fast projectiles, small sprites).
- Genuinely differentiates the algorithms.

Research Question

Given identical preprocessing and matched training budgets, do DQN and PPO achieve statistically equivalent performance, or does one paradigm provide a clear advantage?

The Environment: ALE/KungFuMaster-v5

Atari 2600 beat-em-up exposed through the Arcade Learning Environment (ALE) and the Gymnasium API.

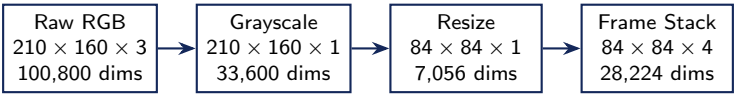
Game Overview

- **Objective:** fight through the Devil's Temple to rescue Sylvia from Mr. X.
- **Enemies:** Grippers, Knife Throwers, Tom Toms, snakes, dragons, falling balls.
- **Scoring:**
  - Standard enemy: $+100$ to $+200$
  - Boss defeat: $+2000$
- **Lives:** 3 per evaluation episode; we terminate after the first death during training.

Technical Specifications

- **Observation:** $210 \times 160 \times 3$ RGB
- **Action space:** `Discrete(14)`
- **Sticky actions:** default 0.25, we set to 0.0
- **Frame skip:** 4, max-pool last 2

MDP Formulation: State Space



Nature of the State

- High-dimensional pixel input (`uint8`, effectively continuous for learning).
- Same protocol as Mnih et al. (2015) — standard Atari practice.
- `uint8` kept for memory efficiency ($\sim$4$\times$ savings vs `float32`); converted to float and normalized inside the network.

Frame Stacking

- A single frame is non-Markovian: velocity and projectile direction are hidden.
- Stacking the last 4 frames **restores the Markov property** via temporal differences.
- Cost: 4$\times$ memory and compute per forward pass.

MDP Formulation: Action Space

`Discrete(14)` — one of 14 actions per timestep

The 14 actions

ID	Action	ID	Action
0	NOOP	7	RIGHTFIRE
1	UP	8	LEFTFIRE
2	RIGHT	9	DOWNFIRE
3	LEFT	10	UPRIGHTFIRE
4	DOWN	11	UPLEFTFIRE
5	DOWNRIGHT	12	DOWNRIGHTFIRE
6	DOWNLEFT	13	DOWNLEFTFIRE

Implications for Algorithm Choice

- Discrete action space $\Rightarrow$ $Q(s, a)$ has finite outputs $\Rightarrow$ value-based methods (DQN) directly applicable via $\pi(s) = \arg\max_a Q(s, a)$.
- **Both algorithms valid here:**
  - DQN: requires discrete actions
  - PPO: handles both discrete and continuous
- If actions were continuous, DQN would not apply directly — DDPG/SAC/TD3 would be required.

A discrete action space lets us compare value-based and policy-based methods without confounding from action-encoding differences.

Reward Structure and Stochasticity

Reward Function (sparse, event-driven)

Event	Reward
Standard enemy (Tom Tom, Knife Thrower)	+100
Tougher enemy (Gripper)	+200
Floor boss defeat	+2000
Damage / movement / idle	0

Stochasticity Sources

Source	Default	Ours	Rationale
`repeat_action_probability`	0.25	0.0	Cleaner comparison; less noise
`noop_max` on reset	30	30	Initial-state diversity
Enemy spawn timing	semi-det.	semi-det.	Inherent ALE behavior

Reward Hacking Risks

- **Minion farming:** waves of 3 enemies spawn endlessly (+600/wave for trivial behavior).
- **Crouching defense:** corner agent kicks attackers, never advances.
- **Score $\neq$ progress:** running forward + boss kill earns $\sim$2800–3500. A passive minion-farmer can score much higher *without solving the level.*

Mitigation: custom reward shaping (Slide 13) explicitly rewards milestone progression.

DQN: Mathematical Foundations

1. The Q-Function (Action-Value)

$$Q^{\pi}(s,a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,\middle|\, s_0 = s, a_0 = a, \pi\right]$$

2. Bellman Optimality Equation

$$Q^{*}(s,a) = \mathbb{E}_{s'}\left[r + \gamma \max_{a'} Q^{*}(s',a') \,\middle|\, s,a\right]$$

3. Deep Q-Learning Update

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\left(y - Q_\theta(s,a)\right)^2\right], \quad y = r + \gamma \max_{a'} Q_{\theta^-}(s',a')$$

$\mathcal{D}$: replay buffer. θ^-: slowly-updated target network. We use **Huber loss** for robustness.

Key Properties

- **Off-policy** — Learns from any past experience $\Rightarrow$ enables replay buffer.
- **Value-based** — Policy is implicit: $\pi(s) = \arg\max_a Q_\theta(s,a)$.
- **Bootstrapping** — Targets depend on Q_{θ^-} itself $\Rightarrow$ requires stabilizers.

DQN: Architecture and Implementation

Network Architecture (mirrors Mnih et al., 2015)

Layer	Type	Filters/Units	Kernel	Stride	Activation
Conv1	2D Conv	32	8×8	4	ReLU
Conv2	2D Conv	64	4×4	2	ReLU
Conv3	2D Conv	64	3×3	1	ReLU
FC1	Linear	512	—	—	ReLU
FC2 (output)	Linear	14	—	—	Linear (Q-values)

Stability Mechanisms

- **Target network** Q_{θ^-}, hard sync every **2,500 steps**.
- **Custom circular replay buffer:** stores single 84×84 frames (not 4-frame tensors); 4-frame stacks reconstructed *on-the-fly* from buffer indices.
- 10% extra slots pre-allocated; transitions whose frames get overwritten are popped. 3M-transition buffer occupies ∼21 GB RAM.
- **Double DQN**; uniform sampling; batch size 128.

Two-Phase ε-Decay

- Phase 1: $1.0 \rightarrow 0.10$ over 2M steps (broad exploration).
- Phase 2: $0.10 \rightarrow 0.01$ over 8M steps (fine-tuning).
- Held at 0.01 thereafter.

Training Hyperparameters

- Adam, $\eta = 1\text{e}{-4}$, $\gamma = 0.99$.
- Huber loss; train every 4 steps; warm-up 80k steps.
- Configured: 15M timesteps; actual run: 3M (stopped for time budget).

PPO: Mathematical Foundations

1. The Policy and Objective

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_t \gamma^t r_t\right], \quad \nabla_\theta J(\theta) = \mathbb{E}_{(s,a)\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)\cdot \hat{A}^\pi(s,a)\right]$$

Advantage: $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$.

2. Clipped Surrogate Objective

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\right)\right]$$

Clip range $\varepsilon = 0.1$. The `min` creates a *pessimistic* lower bound preventing destructive updates.

3. Total Loss (Actor-Critic)

$$L(\theta) = L^{\text{CLIP}}(\theta) - c_1 L^{\text{VF}}(\theta) + c_2 S[\pi_\theta]$$

L^{VF}: value MSE. $S[\pi_\theta]$: entropy bonus ($c_1 = 0.5$, $c_2 = 0.01$). GAE $\lambda = 0.95$ (SB3 default).
Key Properties: On-policy, policy-based, clipped trust-region (no expensive KL).

PPO: Architecture and Implementation

Network Architecture (shared CNN trunk + two heads, via SB3 `CnnPolicy`)

Layer	Type	Filters/Units	Kernel	Stride	Activation
Conv1	2D Conv	32	8×8	4	ReLU
Conv2	2D Conv	64	4×4	2	ReLU
Conv3	2D Conv	64	3×3	1	ReLU
Shared FC	Linear	512	—	—	ReLU
Policy head	Linear	14	—	—	Softmax
Value head	Linear	1	—	—	Linear

On-Policy Rollouts

- `SubprocVecEnv` $\times$ **8 envs** (process-level parallelism, different per-env seeds).
- Rollout: $128 \times 8 = 1024$ steps/iteration.
- 4 epochs per rollout, batch size 256.
- **No replay buffer.**

Added Engineering Beyond SB3

- `TrueScoreWrapper`: raw game score tracking.
- PPO-specific reward shaping (Slide 14).
- Custom CSV callback compatible with DQN's schema.
- Resume checkpointing every 100k steps.
- Periodic in-training video recording.

Hyperparameters: $\eta = 2.5\text{e}{-4}$, $\gamma = 0.99$, *clip* $\varepsilon = 0.1$, *ent_coef 0.01, vf_coef 0.5, max_grad_norm 0.5.*

Custom Preprocessing Pipeline

Standard Atari Preprocessing

- Random no-op reset $\in [1, 30]$ — anti-overfitting.
- Frame skip $= 4$, max-pool last 2 — recovers flickering sprites.
- Grayscale + resize to 84×84 (cv2 area interpolation).
- **UI mask**: zero rows 0–36 (score/timer) and 65+ (floor).
- Stack 4 processed frames as channels.

Knife Dilation (custom)

Problem: knife sprites are 1–2 px at 84×84 — invisible to the CNN.

Fix: identify by RGB (74,74,74), dilate with 3×3 kernel, paint white *before* downsizing.

```
mask = cv2.inRange(play_area,
                   knife_color,
                   knife_color)
dilated = cv2.dilate(mask, kernel)
play_area[dilated > 0] = [255, 255, 255]
```

Impact: knives survive downsampling — agent learns to dodge.

Reward Engineering: Custom Shaping

Base reward is sparse (Slide 7). We add a dense, multi-component signal computed from **RAM-derived game state**.

Shaped Reward Formula

$$r_{\text{shaped}} = r_{\text{step}} + \frac{r_{\text{raw}}}{\text{scale}} + r_{\text{milestone}} + r_{\text{boss\_dmg}} - r_{\text{health}} - r_{\text{death}} + r_{\text{clear}}$$

Eight Components (DQN values)

Component	Value
step_penalty	−0.005
scale_factor	1000.0
base_explore_reward	+0.5
milestone_bonus	+0.2
health_penalty	−0.1
death_penalty	−3.0
level_clear_reward	+25.0
boss_dmg_reward	+0.1

RAM Bytes Used

verified by manual play + memory injection (A. Zanin)

Byte	Meaning
6	Corridor position (0–12)
29	Lives remaining (3–0)
75	Player health (39–0)
76	Boss HP (39–0)

RAM for Reward, Not Observation

Agent input = pixels only.
RAM = reward-designer's tool, equivalent to any human-designed reward.

Why Reward Shaping Must Differ: DQN vs PPO

Component	DQN	PPO	Change
`step_penalty`	-0.005	0.0	**Disabled for PPO**
`milestone_bonus`	$+0.2$	0.0	Disabled for PPO
`health_penalty`	-0.1	-0.02	$5\times$ softer
`death_penalty`	-3.0	-1.0	$3\times$ softer
`level_clear_reward`	$+25.0$	$+5.0$	$5\times$ softer
`base_explore_reward`	$+0.5$	$+0.5$	Same
`boss_dmg_reward`	$+0.1$	$+0.1$	Same
`scale_factor`	1000	1000	Same

Empirical finding: `step_penalty` collapses PPO

With `step_penalty` $= -0.005$ (works perfectly for DQN), PPO's policy collapsed to `NOOP` — the agent learned to NOT MOVE within $\sim$300k steps.

DQN interprets `step_penalty` as:

- Uniform downward shift of all Q-values.
- $\arg\max_a Q$ unchanged $\Rightarrow$ behavior preserved.
- Mild urgency signal $\Rightarrow$ works as intended.

PPO interprets `step_penalty` as:

- All sampled actions get systematically negative $\hat{A}_t$.
- Policy gradient pushes *away* from every action.
- Entropy bonus cannot recover $\Rightarrow$ policy collapses.

Training Setup and Computational Resources

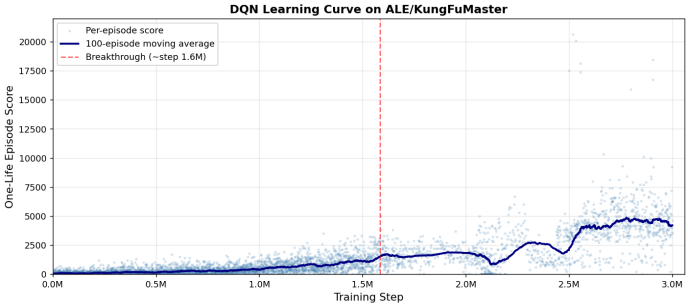
Hardware

Resource	Role
Local workstation (64 GB RAM)	DQN training (large replay buffer)
Google Colab Pro T4	PPO training, development, debugging

Parameter	**DQN**	**PPO**
Optimizer	Adam	Adam
Learning rate	1e−4	2.5e−4
Discount γ	0.99	0.99
Batch size	128	256
Loss	Huber	Clipped surrogate + value MSE + entropy
Grad clip (max-norm)	—	0.5
Replay buffer	3M (custom circular, ∼21 GB)	— (on-policy)
Target network sync	2,500 steps	—
Train frequency	every 4 steps	—
Warm-up (learning_starts)	80,000 steps	—
ε-decay	two-phase 1.0 → 0.10 (2M) → 0.01 (+8M)	—
Parallel envs / per-env seeds	1	8 (different seeds per env)
Rollout length	—	1024 (128×8)
Epochs per rollout	—	4
Clip range ε	—	0.1
GAE λ	—	0.95 (SB3 default)
Entropy coef	—	0.01
Value coef	—	0.5
Configured timesteps	15,000,000	15,000,000
Actual training	3,000,000 (stopped for time budget)	3,000,000 (same)

Same step budget ⇒ fair sample-efficiency comparison; PPO's wall-clock advantage comes from parallelism, not algorithm.

DQN Training Results



Key Statistics

Actual training steps	3M
Total episodes	5,667
Max training (1-life)	20,620
Mean eval score*	**21,916**
Std dev (eval)	5,429.60
Max eval score	**27,500**
Breakthrough step	$\sim$1.6M
ε at breakthrough	$\approx$0.28

**Eval = full 3-lives game; training terminated after first death.*

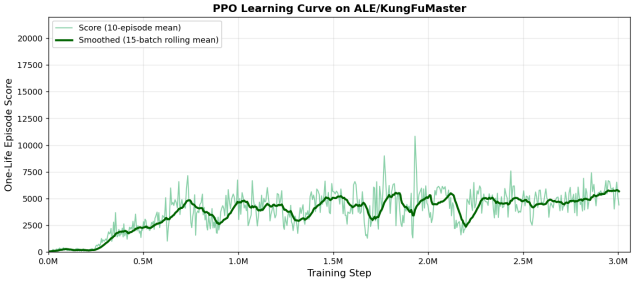
▶ **Watch DQN play**

Three Phases of Learning

Phase	Episodes	ε range	Raw score	Behavior
Random exploration	0–2000	1.0 $\rightarrow$ 0.76	0–300	Random actions; quick deaths
Slow learning	2000–4500	0.76 $\rightarrow$ 0.28	300–2000	Basic combat; occasional level clears
Competent play	4500+	0.28 $\rightarrow$ 0.09	2000–20,620	Reliably clears level 1

Breakthrough at step $\sim$1.6M: 100-episode rolling score climbs past 1,500 and continues upward to $\sim$5,000 by 3M.

PPO Training Results



Key Statistics

Actual training steps	3M
Parallel envs	8 (SubprocVecEnv)
Cumulative episodes	∼5,865
Max training (1-life)	10,826
Mean eval score*	**10,312**
Std dev (eval)	5,108.45
Max eval score	**19,000**

**Eval = full 3-lives game.*

▶ **Watch PPO play**

PPO Learning Trajectory

Phase	Approx. steps	Behavior
Initial exploration	0–300k	High-entropy stochastic policy; low scores
Rapid improvement	300k–1M	Commits to combat; one-life score climbs to ∼5,000
Plateau / refinement	1M–3M	Small oscillating gains; training stopped at 3M

Algorithm-specific observations

PPO's curve is smoother than DQN's stepwise breakthrough. PPO plateaued earlier; matching DQN's growth requires training beyond 3M.

Statistical Significance of the Comparison

Evaluation protocol: each trained policy played **50 evaluation games** with **50 different environment seeds** (deterministic action selection, 3 lives per game).

Descriptive statistics

Metric	DQN	PPO
Mean eval score	21,916.00	10,312.00
Std deviation	5,429.60	5,108.45
Max observed	27,500	19,000
Mean difference (DQN − PPO)	+11,604	

Hypothesis tests

Test	DQN dist.	PPO dist.	p-value
Shapiro–Wilk (normality)	NOT normal	NOT normal	DQN 3.16e−9 / PPO 3.24e−8
Welch's t-test *(parametric)*	—	—	1.46e−18
Mann–Whitney U *(non-param.)*	—	—	**2.10e−11**

Interpretation

Non-normal distributions $\Rightarrow$ **Mann–Whitney U is the valid test**. $p = 2.10\text{e}{-11}$: probability the gap arose by chance is $\sim$1 in 50 billion. **DQN significantly outperforms PPO**.

Failure Analysis and Honest Limitations

Failure modes and experimental issues

- Reward hacking risks (identified during reward design — Slide 7): minion farming and crouching defense are theoretically incentivized by the sparse base reward; mitigated through custom shaping but not provably eliminated.
- Neither algorithm progressed beyond floor 1: agents repeatedly defeat the floor-1 boss but no transition to a higher floor was observed (the emulator does not appear to expose the floor-progression mechanic of the original cabinet — under investigation).
- PPO policy sensitivity: collapsed with DQN-default `step_penalty` (see Slide 14) — requires algorithm-aware reward design.

Algorithmic limitations

- **DQN:** "deadly triad" (function approximation + bootstrapping + off-policy) — no convergence guarantee under deep RL.
- **PPO:** on-policy = sample-inefficient; smoother but lower performance — consistent with literature on sparse-reward Atari.

Methodological limitations

- Single training seed (42) per algorithm; PPO's 8 envs each use different per-env seeds, providing some training-time variability.
- Generalization assessed via 50 different evaluation seeds.
- `repeat_action_probability = 0` — not standard ALE benchmark.
- PPO hyperparameters at SB3 defaults.
- Training stopped at 3M of planned 15M steps for time budget.
- No ablation studies.
- RAM used for reward computation only (Slide 13).

Takeaway

Our agents play KungFuMaster competently on floor 1 but do not solve the game. The DQN vs PPO comparison is valid *under our experimental conditions*; absolute scores reflect those conditions, not algorithmic ceilings.

Conclusion and References

Key findings

1. **DQN significantly outperforms PPO** on KungFuMaster under matched conditions. Mean: 21,916 vs 10,312; Mann–Whitney $p < 10^{-10}$.
2. **Algorithm-specific reward shaping is essential.** Same reward function, different parameter values: on-policy and off-policy methods do not interpret rewards identically.
3. **Game-specific preprocessing matters.** Custom knife dilation closed a perception gap.
4. Both algorithms reached competent floor-1 play; **neither progressed further**.

Future work

- Multi-seed training (beyond per-env PPO seeds) with confidence intervals on training variance.
- Extended budget (full 15M+ steps); Rainbow DQN; PPO + RND.
- Ablation studies on reward components; investigation of floor-progression mechanics in the ALE emulator.

Code and full training results

`https://github.com/AndreaZanin02/KungFuMaster`

References

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Thank you. Questions?